Phishing Website Detection

Random Forest | Binary Classification

Student Name: _____

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Machine Learning Project Report

Complete personal details and insert the exact evaluation scores printed by your Jupyter Notebook before submission.

1. Introduction

This project builds a machine learning classifier to identify phishing websites. It uses the **Phishing_Legitimate_full.csv** dataset and the target column **CLASS_LABEL**, where 1 represents phishing and 0 represents legitimate websites.

2. Problem statement

Phishing websites imitate trusted services to steal credentials or personal information. The goal is to classify a website using its measured URL and page-related features. This is a binary classification problem.

3. Dataset overview

The dataset contains numeric indicators describing website or URL characteristics and a class label. Confirm the exact dimensions, feature names and class counts using df.shape, df.info() and CLASS_LABEL.value_counts() in Jupyter.

4. Project workflow

• Load the CSV using pandas.

• Inspect data types, missing values and duplicate records.

• Remove duplicate rows and fill missing values if needed.

• Drop the id column because it is an identifier, not a meaningful website signal.

• Keep CLASS_LABEL as the target and separate it from the input features.

• Split the data into training and testing sets with stratification.

5. Data cleaning

Remove duplicate rows. Check for missing values and fill numeric columns with their median where appropriate. Drop the id field as directed by the project guide. Do not remove legitimate predictive features unless there is a clear reason.

6. Exploratory data analysis

• Plot counts of legitimate and phishing examples.

• Inspect histograms of numeric website features.

• Use box plots to compare selected features by CLASS_LABEL.

• Create a correlation heatmap to identify strongly related features.

• Record five observations that are supported by the charts.

7. Model: Random Forest

Random Forest is an ensemble classifier that combines predictions from multiple decision trees. Each tree is trained using a sample of the training data and a random subset of features. Aggregating many trees generally reduces the instability of a single decision tree. Feature scaling is not required.

8. Feature importance

After fitting the model, inspect feature_importances_ and rank the strongest predictors. Explain the top five features using their actual column names and values from the notebook. Feature importance describes model reliance, not proof of causation.

9. Evaluation

Evaluate the test set with accuracy, precision, recall, F1-score, specificity and ROC-AUC. For this security task, recall for the phishing class is particularly important because false negatives are phishing sites incorrectly classified as legitimate.

Evaluation results

Metric	Notebook result
Accuracy	_____
Precision	_____
Recall	_____
F1-score	_____
Specificity	_____
ROC-AUC	_____

10. Model tuning

Compare Random Forest settings using different values of n_estimators and max_depth. Add each experiment to the results list. Compare the metrics and feature-importance plots, then select the final model with attention to phishing recall.

11. Conclusion and limitations

The project demonstrates how website characteristics can be used to flag potential phishing pages. A model can make errors when websites change or when attackers use new techniques. It should support, not replace, security checks, browser protections and human review. Report the exact test metrics from the notebook.